UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Problems on Euler's Theorem

Complete study pack for class 006: testing homogeneity with the scaling test, verifying Euler's theorem, first- and second-order problems, and the extension when f(u) is homogeneous

Compiled from the PESU class 006 slide deck, its two AV-summary transcripts, the Unit 1 class-work sheet and past ESA papers. Generated 14 October 2026.

· How to use this pack

Topic 1 tests whether a function is homogeneous; topic 2 verifies Euler's theorem by differentiating; topics 3 and 4 are the two kinds of exam problem: u itself homogeneous, or some $f(u)$ homogeneous. The theory (definitions, proofs, applications) is in the class 005 pack "Homogeneous Functions and Euler's Theorem"; the statements are repeated here only so that each problem can be solved from this pack alone. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S006·12	Deck 006, slide 12 (rendered .pptx)
T006a 00:05:18	Transcript 006 part I (slides 3–4, 9.1 min)
T006b 00:08:20	Transcript 006 part II (slides 5–17, 12.9 min)
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

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1. 1 · Testing homogeneity: the scaling test
2. 2 · Verifying Euler's theorem directly
3. 3 · When u itself is homogeneous: first and second order
4. 4 · When f(u) is homogeneous: the extension
5. Appendix A · Symbol glossary
6. Appendix B · Sources, coverage and gaps

1 Testing homogeneity: the scaling test

Before Euler's theorem can be used, you must show the function is homogeneous and find its degree. Class 006 does this with the scaling test: replace x by λx and y by λy; if the result is $\lambda^n u(x,y)$, then u is homogeneous of degree n. The degree can be positive, negative or zero; if no power of λ factors out, the function is not homogeneous.

Key terms and definitions

Homogeneous function (scaling form). $u(x,y)$ is homogeneous of degree n if $u(\lambda x, \lambda y) = \lambda^n\, u(x,y)$. Here n can be positive, negative or even zero. S006·5 · T006b 00:00:52 · T006b 00:01:11

Not homogeneous. u is not homogeneous when $u(\lambda x, \lambda y) \neq \lambda^n u(x,y)$ for every n: no single power of λ can be taken out. S006·8 · T006b 00:04:59

Formulas

E1.1 · Scaling test ESA ★★

$$u(\lambda x, \lambda y) = \lambda^n\, u(x,y) \implies u \text{ homogeneous of degree } n$$

gives	Whether u is homogeneous, and its degree.
applies	Any function; with three variables scale z too.
note	If no power λ^n factors out, u is not homogeneous.

S006·5 · T006b 00:00:52

E1.2 · Degree of a quotient ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q$$

gives	The degree without writing out λ.
applies	P, Q homogeneous.
note	$\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $1-4=-3$.

added for clarity S006·4 · T006a 00:06:10

Graphs and diagrams

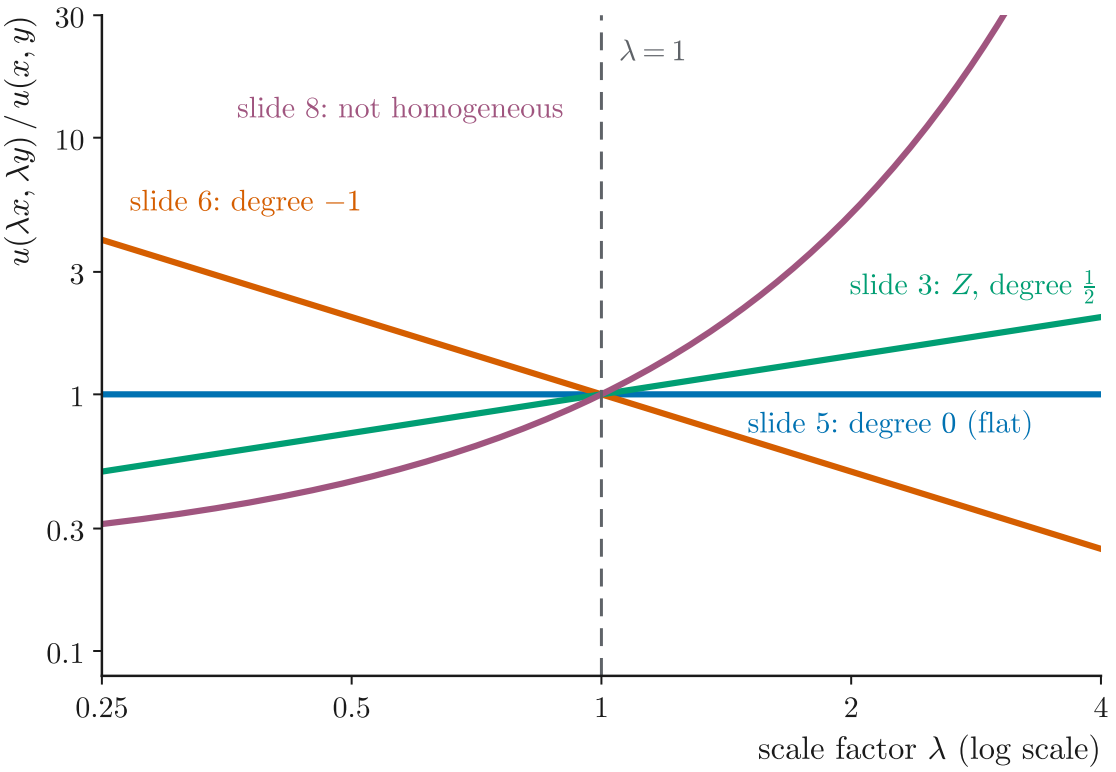
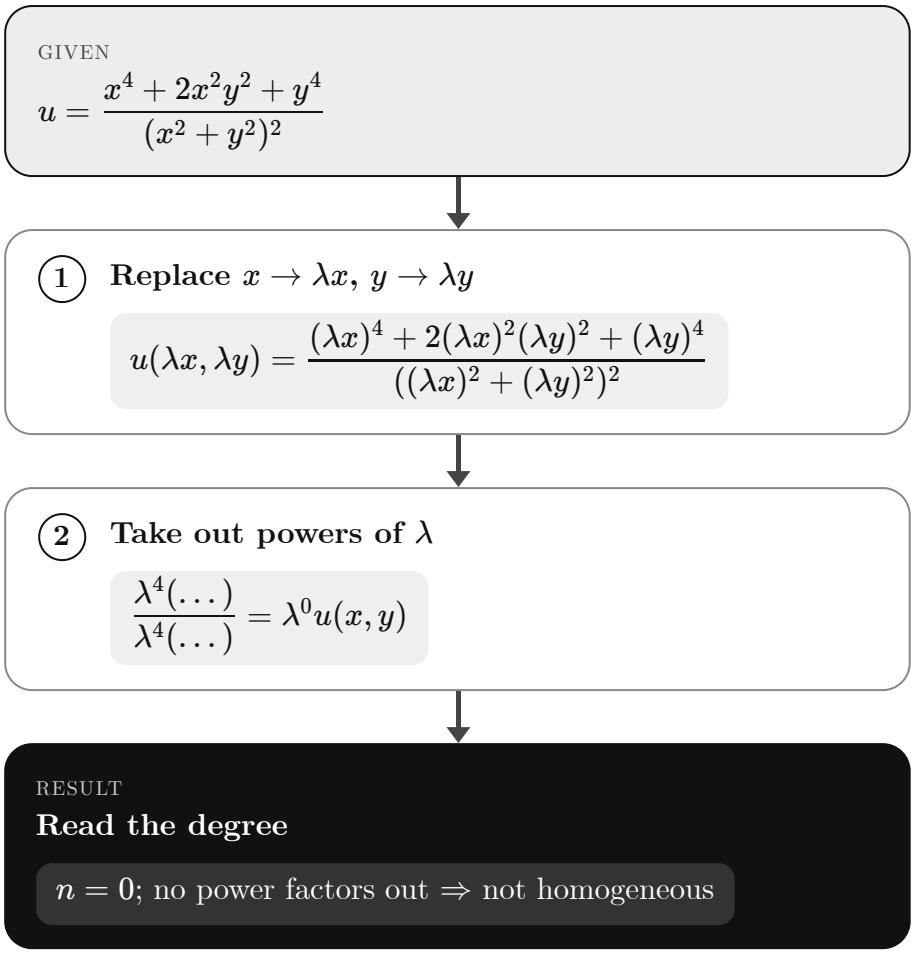


Figure G1.1. The scaling test on log-log axes. ESA ★★

ADDED FOR CLARITY S006·3 · S006·5 · S006·6 · S006·8

How to read it. Each curve is $u(\lambda x, \lambda y)/u(x, y)$ for a deck function, plotted against λ on log-log axes. A homogeneous function gives λ^n: a straight line of slope n through $(1, 1)$. Slide 006·5 is flat (degree 0), slide 006·6 falls with slope -1, Z from slide 006·3 rises with slope $\frac{1}{2}$. Slide 006·8's function curves away: it is not homogeneous. Base point $(1, 2)$; $(2, 1)$ for slide 006·6, where $\sin^{-1}(y/x)$ needs $y \leq x$.

Flowcharts



Remark. Use the test on u (topic 3) or on $f(u)$ (topic 4); the steps are the same.

Figure F1.1. The scaling test ESA ★★

S006·5 · T006b 00:01:41 · T006b 00:02:37

Important points and why

1. Write $u(\lambda x, \lambda y)$, take the highest power of λ out of the numerator and the denominator, and cancel. ESA ★★ S006·5 · T006b 00:02:17
2. A λ inside a ratio cancels: $\sin^{-1}\frac{\lambda y}{\lambda x} = \sin^{-1}\frac{y}{x}$, so functions of y/x add degree 0. ESA ★★ S006·6 · T006b 00:03:34
3. $e^{\lambda x}$ cannot be written as $\lambda^n e^x$, so exponentials of x or y alone break homogeneity (slide 006·8). ESA ★★ S006·8 · T006b 00:04:40
4. Simplify first when you can: slide 006·5 is $\frac{(x^2+y^2)^2}{(x^2+y^2)^2} = 1$ and slide 006·7 is $\frac{xy(x+y)}{(x+y)^3} = \frac{xy}{(x+y)^2}$; both have degree 0. ESA ★ (added for clarity) S006·5 · S006·7

Why does slide 006·8 fail the test?

$u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda (x - y)}$. The denominator gives λ^{-1}, but $e^{\lambda x} - e^{\lambda y}$ is not a power of λ times $e^x - e^y$. Figure G1.1 shows the ratio is not a straight line on log-log axes.

ADDED FOR CLARITY S006·8

Worked examples

W1.1 · Degree 0 (slide 006·5) ESA ★

GIVEN $u = \dfrac{x^4 + 2x^2y^2 + y^4}{(x^2 + y^2)^2}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda^4 (x^4 + 2x^2y^2 + y^4)}{\lambda^4 (x^2 + y^2)^2} = \lambda^0 u(x, y)$.

ANSWER Homogeneous of degree 0

CHECK The numerator is $(x^2 + y^2)^2$, so $u \equiv 1$: a constant, degree 0.

S006·5 · T006b 00:00:24

W1.2 · Degree −1 (slide 006·6) ESA ★★

GIVEN $u = \dfrac{\sin^{-1}(y/x)}{\sqrt{x^2 + y^2}}$.

STEP 1 $\sin^{-1} \frac{\lambda y}{\lambda x} = \sin^{-1} \frac{y}{x}$: the λ cancels.

STEP 2 $\sqrt{(\lambda x)^2 + (\lambda y)^2} = \lambda \sqrt{x^2 + y^2}$.

STEP 3 $u(\lambda x, \lambda y) = \lambda^{-1} u(x, y)$.

ANSWER Homogeneous of degree -1

CHECK sympy: $u(\lambda x, \lambda y)/u = 1/\lambda$.

S006·6 · T006b 00:03:02

W1.3 · Degree 0 (slide 006·7) ESA ★

GIVEN $u = \dfrac{x^2 y + x y^2}{(x + y)^3}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda^3 (x^2 y + x y^2)}{\lambda^3 (x + y)^3} = \lambda^0 u$.

ANSWER Homogeneous of degree 0

CHECK $u = \frac{xy}{(x+y)^2}$, a function of y/x only.

S006·7 · T006b 00:04:06

W1.4 · Not homogeneous (slide 006·8) ESA ★★

GIVEN $u = \dfrac{e^x - e^y}{x - y}.$

STEP 1 $u(\lambda x, \lambda y) = \dfrac{e^{\lambda x} - e^{\lambda y}}{\lambda (x - y)}.$

STEP 2 No factor λ^n can be taken out of $e^{\lambda x} - e^{\lambda y}$.

ANSWER u is not homogeneous

CHECK Figure G1.1: the ratio curves on log-log axes.

S006·8 · T006b 00:04:40

SHORT-NOTE RECAP

Scaling test: $u(\lambda x, \lambda y) = \lambda^n u(x, y)$ means degree n (positive, negative or zero). Slide examples: $\frac{x^4+2x^2y^2+y^4}{(x^2+y^2)^2}$ degree 0; $\frac{\sin^{-1}(y/x)}{\sqrt{x^2+y^2}}$ degree -1; $\frac{x^2y+xy^2}{(x+y)^3}$ degree 0; $\frac{e^x-e^y}{x-y}$ not homogeneous.

2 Verifying Euler's theorem directly

To verify the theorem, compute both sides separately: find the degree n, then differentiate, form $xu_x + yu_y$ and simplify until it equals nu. Slide 006·3 does this for $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ with the quotient rule. It is long; in every other problem the theorem is applied, not verified.

Key terms and definitions

> *"Euler's theorem: if $u = f(x, y)$ is a homogeneous function of degree n, then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."*
>
> S006·10 · T006a 00:00:00

Formulas

E2.1 · Euler's theorem ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

gives	The first-order Euler sum.
applies	u homogeneous of degree n; add $z\,u_z$ for three variables.
note	To verify, compute the left side by differentiating and show it equals nu.

S006·3 · S006·10 · T006a 00:00:00

Important points and why

1. Verification is a check of the theorem, not a method: the lecture spends four minutes on slide 006·3; applying the theorem takes one line. ESA ★ T006a 00:04:57 · T006a 00:08:41
2. In slide 006·3, group the terms with the common denominator $(\sqrt{x} + \sqrt{y})^2$ and cancel one factor of $\sqrt{x} + \sqrt{y}$. ESA ★ T006a 00:02:59 · T006a 00:03:17

Exam trap. Slide 006·3 says "verify for $\cos^{-1}(\dots)$" but verifies for $Z = \cos u$. For u itself, $xu_x + yu_y = \frac{1}{2}\frac{\cos u}{-\sin u} = -\frac{1}{2}\cot u$, not $\frac{1}{2}u$. ADDED FOR CLARITY S006·3

Worked examples

W2.1 · Verify Euler's theorem (slide 006·3) ESA ★★

GIVEN $u = \cos^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$; verify Euler's theorem for $Z = \cos u = \dfrac{x+y}{\sqrt{x}+\sqrt{y}}$.

STEP 1 $Z = \dfrac{x(1+y/x)}{\sqrt{x}(1+\sqrt{y/x})} = \sqrt{x}\, f\left(\frac{y}{x}\right)$: degree $\frac{1}{2}$, so Euler predicts $xZ_x + yZ_y = \frac{1}{2}Z$.

STEP 2 Quotient rule: $Z_x = \dfrac{(\sqrt{x}+\sqrt{y}) - \frac{x+y}{2\sqrt{x}}}{(\sqrt{x}+\sqrt{y})^2}$, so $xZ_x = \dfrac{x\sqrt{x} + 2x\sqrt{y} - y\sqrt{x}}{2(\sqrt{x}+\sqrt{y})^2}$.

STEP 3 By symmetry $yZ_y = \dfrac{y\sqrt{y} + 2y\sqrt{x} - x\sqrt{y}}{2(\sqrt{x}+\sqrt{y})^2}$.

STEP 4 Add: numerator $x\sqrt{x} + x\sqrt{y} + y\sqrt{x} + y\sqrt{y} = (x+y)(\sqrt{x}+\sqrt{y})$, so the sum is $\dfrac{x+y}{2(\sqrt{x}+\sqrt{y})} = \frac{1}{2}Z$.

ANSWER $\boxed{xZ_x + yZ_y = \frac{1}{2}Z}$: verified

CHECK sympy: residual 0. For u itself: $xu_x + yu_y = -\frac{1}{2}\cot u$ (added for clarity).

S006·3 · T006a 00:00:18 · T006a 00:04:11

SHORT-NOTE RECAP

To verify: find n, differentiate, form $xu_x + yu_y$, simplify to nu. For $Z = \frac{x+y}{\sqrt{x}+\sqrt{y}}$ ($n = \frac{1}{2}$): $xZ_x + yZ_y = \frac{(x+y)(\sqrt{x}+\sqrt{y})}{2(\sqrt{x}+\sqrt{y})^2} = \frac{1}{2}Z$.

3 When u itself is homogeneous: first and second order

If u is homogeneous of degree n, write the answers down: $xu_x + yu_y = nu$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$; their sum is n^2u. For a sum of parts with different degrees, apply the result to each part with its own degree. Most ESA questions on Euler's theorem are of this type.

Key terms and definitions

> *"By Euler's theorem for a homogeneous function of degree n: $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$."* S006·11

Formulas

E3.1 · Second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = n(n-1)\,u$$

GIVES	The second-order Euler sum.
APPLIES	u homogeneous of degree n.
NOTE	$n=-1$ gives $2u$; $n=-\frac{2}{3}$ gives $\frac{10}{9}u$.

S006·11 · T006b 00:07:23

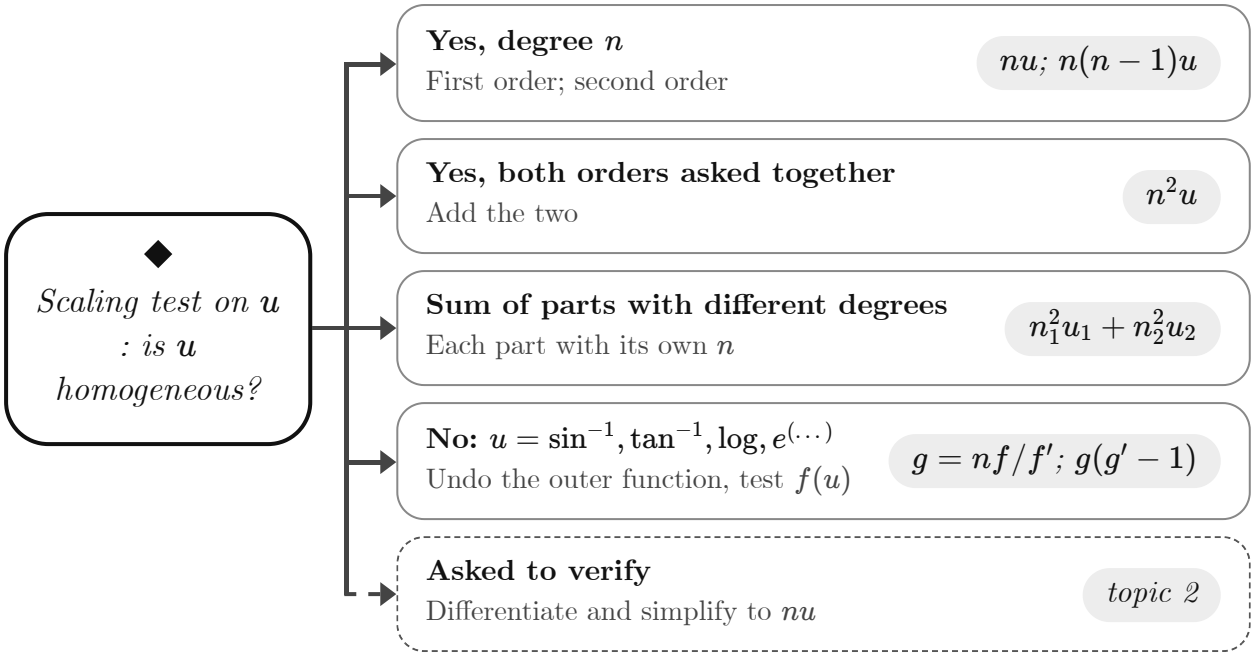
E3.2 · First plus second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} + x\,u_x + y\,u_y = n^2u$$

GIVES	The combination asked in ESA Jan 2024 sets 3 and 4.
APPLIES	u homogeneous of degree n; for u_1+u_2 use $n_1^2u_1 + n_2^2u_2$.
NOTE	$n(n-1)+n = n^2$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

Flowcharts



Remark. Exam questions say "using Euler's theorem": always state the homogeneity and the degree first.

Figure F3.1. Which Euler formula? ESA ★★★

T006b 00:06:11 · T006b 00:09:08 · S006·13

Important points and why

1. First prove that u is homogeneous and find n; then write both answers without differentiating. ESA ★★★ T006b 00:06:11 · T006b 00:06:45
2. Give the answer in terms of x and y too: $-u = -\frac{x-y}{x^2+y^2}$ and $2u = \frac{2(x-y)}{x^2+y^2}$. ESA ★★ S006·10 · S006·11
3. For a sum of parts with different degrees, apply the corollary part by part: $n_1(n_1-1)u_1 + n_2(n_2-1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22

Exam trap. $n = -1$ gives $n(n-1) = (-1)(-2) = +2$, not -2: the second-order answer on slide 006·11 is $+2u$. ADDED FOR CLARITY S006·11

Worked examples

W3.1 · Degree −1, both orders (slides 006·9–11) ESA ★★★

GIVEN $u = \dfrac{x-y}{x^2+y^2}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $u(\lambda x, \lambda y) = \dfrac{\lambda(x-y)}{\lambda^2(x^2+y^2)} = \lambda^{-1}u$: $n = -1$.

STEP 2 (i) Euler: $-1 \cdot u$.

STEP 3 (ii) Corollary: $(-1)(-2)u = 2u$.

ANSWER $\boxed{\text{(i) } -\dfrac{x-y}{x^2+y^2},\ \text{(ii) } \dfrac{2(x-y)}{x^2+y^2}}$

CHECK sympy: both residuals 0.

S006·9 · S006·10 · S006·11 · T006b 00:05:27

W3.2 · Fractional degree (CW) ESA ★★

GIVEN $u = \dfrac{y^{1/3} - x^{1/3}}{x+y}$; find both Euler sums.

STEP 1 Degree $\frac{1}{3} - 1 = -\frac{2}{3}$.

STEP 2 (i) $-\frac{2}{3}u$.

STEP 3 (ii) $\left(-\frac{2}{3}\right)\left(-\frac{5}{3}\right)u = \frac{10}{9}u$.

ANSWER $\boxed{\text{(i) } -\frac{2}{3}u,\ \text{(ii) } \frac{10}{9}u}$

CHECK CW answers agree.

CW U1 P19

W3.3 · Degree 1 with an inverse sine (CW) ESA ★★

GIVEN $u = \sqrt{y^2 - x^2}\,\sin^{-1}(x/y)$; find $xu_x + yu_y$.

STEP 1 $\sqrt{y^2-x^2} = y\sqrt{1-(x/y)^2}$: degree 1; $\sin^{-1}(x/y)$: degree 0.

STEP 2 Degree $1 + 0 = 1$.

ANSWER $\boxed{xu_x + yu_y = u}$

CHECK CW answer: u.

CW U1 P21

W3.4 · Prove 5u and 20u (ESA) ESA ★★★

GIVEN $u = x^3y^2\sin^{-1}(y/x)$. Prove (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

STEP 1 $u(\lambda x, \lambda y) = \lambda^5 x^3y^2\sin^{-1}(y/x)$: degree 5.

STEP 2 (i) $5u$.

STEP 3 (ii) $5 \cdot 4\,u = 20u$.

ANSWER (i) $5u$, (ii) $20u$

CHECK Direct differentiation in sympy agrees.

ESA Jul-23 1a

W3.5 · Sum of two degree-3 parts (CW) ESA ★★

GIVEN $u = x^3\sin^{-1}(y/x) + y^3\tan^{-1}(x/y)$; find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 Both parts have degree 3, so u does.

STEP 2 $3 \cdot 2\,u$.

ANSWER $6u$

CHECK CW answer: $6u$.

CW U1 P22

W3.6 · Degrees 3 and −3 together (ESA) ESA ★★★

GIVEN $u = x^3\sin^{-1}(y/x) + y^{-3}\tan^{-1}(x/y)$; evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$.

STEP 1 u_1: degree 3; $u_2 = y^{-3}\tan^{-1}(x/y)$: degree -3.

STEP 2 First + second order gives n^2: $9u_1 + 9u_2$.

ANSWER $9u$

CHECK CW answer: $9u$; sympy agrees.

ESA Jan-24 s4 1d · CW U1 P23

W3.7 · Degrees 3 and 4, value at (1, 1) (ESA) ESA ★★★

GIVEN $u = x^3 \sin^{-1}(y/x) + x^4 \tan^{-1}(y/x)$; find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $(1, 1)$.

STEP 1 $9u_1 + 16u_2$ with $u_1 = x^3 \sin^{-1}(y/x)$, $u_2 = x^4 \tan^{-1}(y/x)$.

STEP 2 At $(1, 1)$: $u_1 = \frac{\pi}{2}$, $u_2 = \frac{\pi}{4}$.

STEP 3 $\frac{9\pi}{2} + 4\pi$.

ANSWER $\boxed{\dfrac{17\pi}{2}}$

CHECK sympy: the expression equals $9u_1 + 16u_2$ identically.

ESA Jan-24 s3 1c

W3.8 · A function of ratios only (ESA) ESA ★★

GIVEN $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$; find $xu_x + yu_y + zu_z$.

STEP 1 Scaling x, y, z by λ leaves every ratio unchanged: degree 0.

STEP 2 Euler in three variables: $0 \cdot u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = 0}$

CHECK Same as the chain-rule method (sympy).

ESA Jan-24 s1 1d

Past-paper questions

ESA Jul 2023 (6 marks) UE22MA141A Q1.a · solved in W3.4

If $u(x, y) = x^3y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $xu_x + yu_y = 5u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 20u$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.c · solved in W3.7

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.d · solved in W3.6

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3} \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W3.8

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $xu_x + yu_y + zu_z$ (degree 0; also solvable by the chain rule).

SHORT-NOTE RECAP

u homogeneous of degree n: $xu_x + yu_y = nu$, $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = n(n-1)u$, sum n^2u. $\frac{x-y}{x^2+y^2}$: $-u$, $2u$. ESA: $5u$ & $20u$; $9u$; $17\pi/2$; ratios only give 0.

4 When f(u) is homogeneous: the extension

For $u = \sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\cdots)}$, u is not homogeneous, but undoing the outer function gives a homogeneous $f(u)$: $\sin u$, $\tan u$, e^u or $\log u$. If $f(u)$ has degree n, then $xu_x + yu_y = g(u) = n\,f(u)/f'(u)$ and the second-order sum is $g(u)[g'(u) - 1]$. Class 006 works four such problems; ESA May and June 2024 ask the $\tan^{-1}$ case.

Key terms and definitions

Extension function g(u). $g(u) = n\dfrac{f(u)}{f'(u)}$, where $f(u)$ is homogeneous of degree n in x and y.

S006·13 · S006·15 · S006·17

> *"For $f(u)$ a homogeneous function of degree n: $xu_x + yu_y = n\left(\dfrac{f(u)}{f'(u)}\right) = g(u)$, and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$."* S006·13 · S006·15 · S006·17

Formulas

E4.1 · Extension, first order ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

gives	The Euler sum when u is not homogeneous but $f(u)$ is.
applies	$f(u)$ homogeneous of degree n; add $z\,u_z$ for three variables.
note	$\sin \to n\tan u$, $\tan \to \frac{n}{2}\sin 2u$, $e^u \to n$, $\log \to nu\log u$.

S006·13 · S006·15 · S006·17 · T006b 00:09:08

E4.2 · Extension, second order ESA ★★★

$$x^2u_{xx} + 2xy\,u_{xy} + y^2u_{yy} = g(u)\,\big[g'(u) - 1\big]$$

gives	The second-order sum for the extension.
applies	g from E4.1; differentiate g with respect to u.
note	If g is a constant (the $\log$ case), the answer is $-g$.

S006·15 · S006·17 · T006b 00:10:59

E4.3 · Deriving the extension by the chain rule ESA ★★

$$F = f(u):\ xF_x + yF_y = nF, \quad F_x = f'(u)\,u_x \ \Rightarrow\ f'(u)\,(xu_x + yu_y) = nf(u)$$

gives	Where E4.1 comes from; T006a works slide 006·4 this way.
applies	When you prefer not to quote E4.1.
note	On slide 006·4: $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$; divide by $\cos u$.

S006·4 · T006a 00:07:33 · T006a 00:08:22

Graphs and diagrams

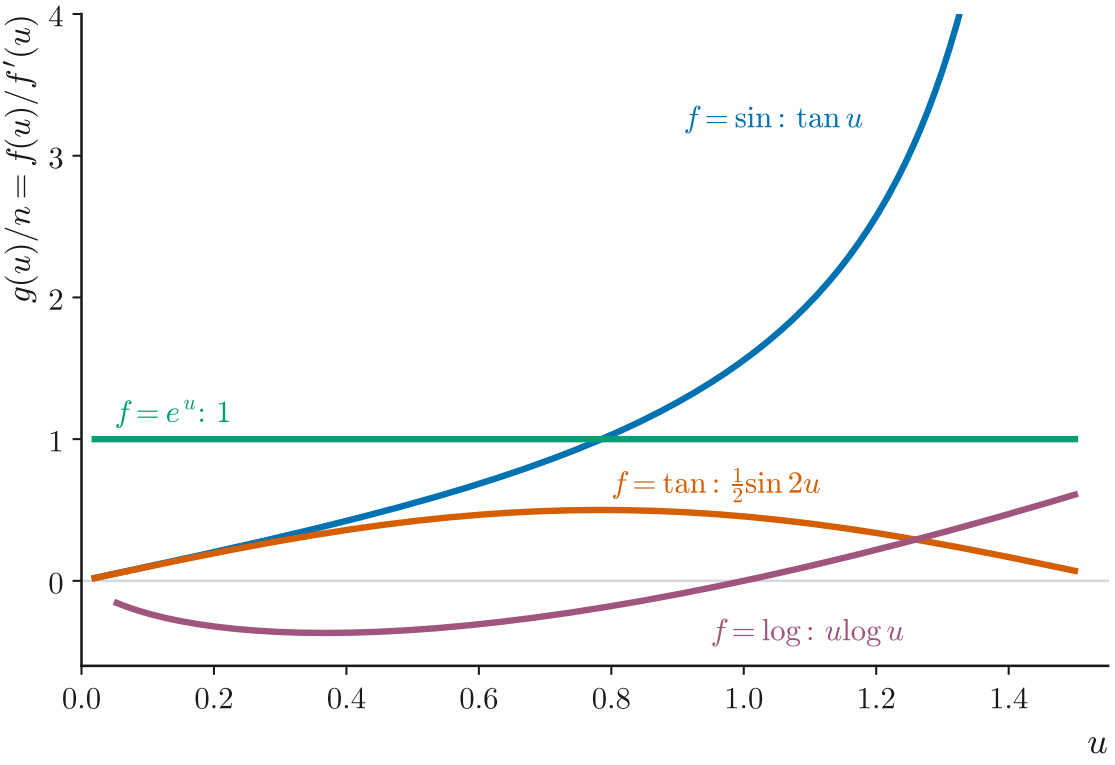
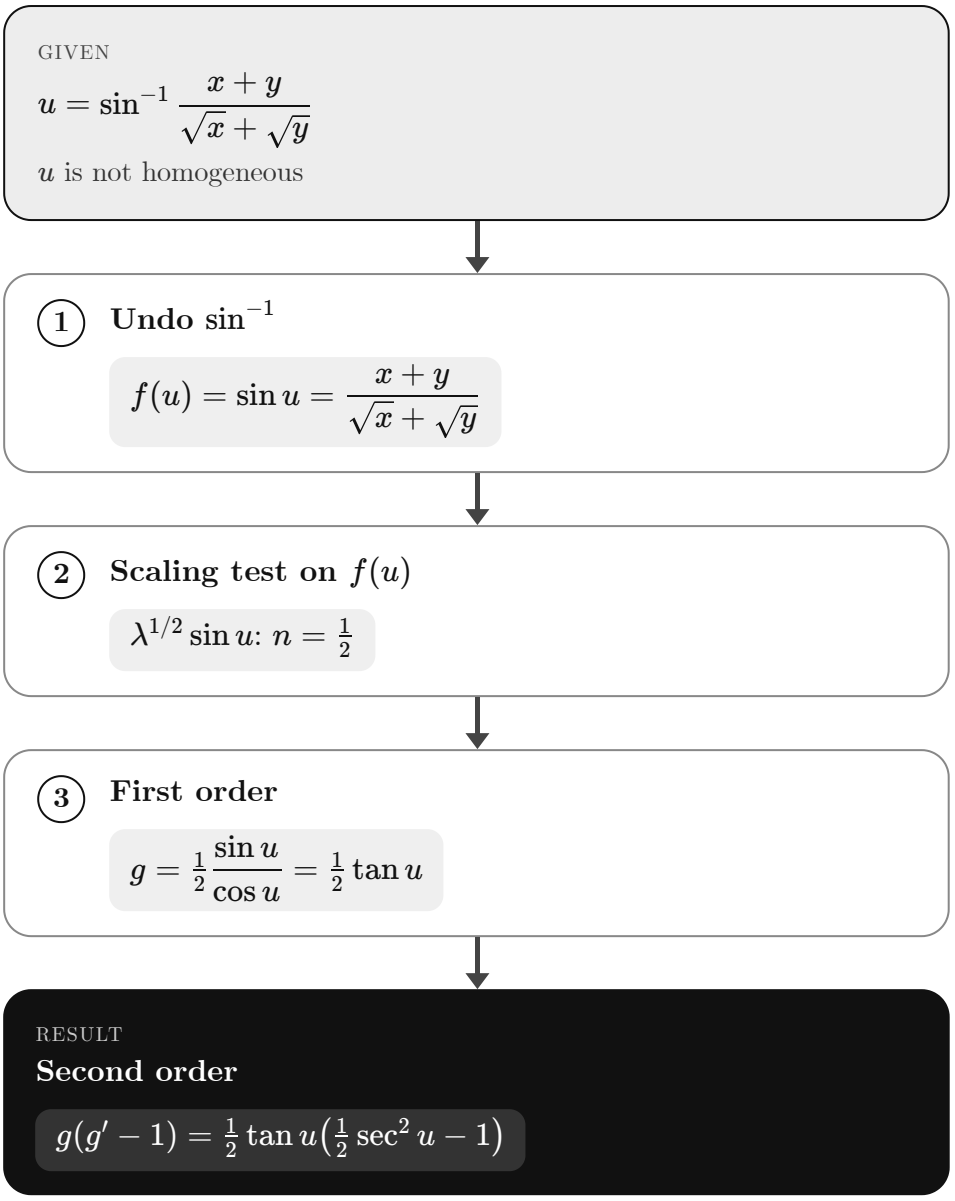


Figure G4.1. The extension function $f(u)/f'(u)$ for each choice of f. ESA ★★

ADDED FOR CLARITY S006·13 · S006·15 · S006·17 · ESA May-24 1a

How to read it. Multiply the curve by the degree n to get $g(u) = xu_x + yu_y$. For $f = \sin$ the sum grows without bound as $u \to \pi/2$; for $f = \tan$ it stays between $\pm\frac{n}{2}$; for $f = e^u$ it is the constant n, so the second order is $-n$; for $f = \log$ it is $nu\log u$, negative for $u < 1$ (on slide 006·12 $u = e^{(\cdots)} > 1$, so it is positive).

Flowcharts



Remark. The same four steps give $4u\log u$ for $e^{(\cdots)}$ (slide 006·12) and 1, -1 for $\log(\dots)$ (slide 006·14).

Figure F4.1. Using the extension (slides 006·16–17) ESA ★★★

S006·16 · S006·17 · T006b 00:11:40

The extension for each outer function

$u =$	$f(u)$	$g(u) = xu_x + yu_y$	$g(g'-1)$	**In class 006**
$\sin^{-1}(P)$	$\sin u$	$n\tan u$	$n\tan u\,(n\sec^2 u - 1)$	$n=-3$ (slide 4); $n=\frac{1}{2}$ (slide 16)
$\cos^{-1}(P)$	$\cos u$	$-n\cot u$	$-n\cot u\,(n\csc^2 u - 1)$	$n=\frac{1}{2}$ (slide 3, u itself)
$\tan^{-1}(P)$	$\tan u$	$\frac{n}{2}\sin 2u$	$\frac{n}{2}\sin 2u\,(n\cos 2u - 1)$	$n=2$ (ESA May/June 2024)
$\log(P)$	e^u	n	$-n$	$n=1$ (slide 14)
e^P	$\log u$	$nu\log u$	$nu\log u\,(n\log u + n - 1)$	$n=4$ (slide 12)

ADDED FOR CLARITY S006·4 · S006·13 · S006·15 · S006·17 · ESA May-24 1a

Important points and why

1. When u is $\sin^{-1}$, $\tan^{-1}$, $\log$ or e of something, apply the inverse function to both sides: that is $f(u)$. ESA ★★★ T006b 00:08:20 · T006b 00:09:55 · T006b 00:11:40
2. The degree belongs to $f(u)$, found by the scaling test on $f(u)$ (λ^4 on slide 006·12, λ on 006·14, $\lambda^{1/2}$ on 006·16). ESA ★★★ S006·12 · S006·14 · S006·16
3. "Whenever you have a complicated function, you can always use Euler's theorem if you feel that the function is homogeneous" — the lecture's closing advice for part I. ESA ★★ T006a 00:08:53
4. Three variables work the same way: on slide 006·4, $\sin u$ has degree -3, so $xu_x + yu_y + zu_z = -3\tan u$. ESA ★★ S006·4 · T006a 00:06:54

Why does the log case give -1 for the second order?
$f(u) = e^u$ gives $g = n\,e^u/e^u = n$, a constant. So $g' = 0$ and $g(g'-1) = -n$; slide 006·14 has $n = 1$.
S006·15

Why can't Euler's theorem be applied to u directly on slides 006·12–17?
u is not homogeneous: for $u = \log\frac{x^2+y^2}{x+y}$, $u(\lambda x, \lambda y) = \log\lambda + u$, which is a shift, not a factor λ^n.
ADDED FOR CLARITY T006b 00:09:55

Exam trap. For $u = e^{(\cdots)}$ the function is $f(u) = \log u$, so $f' = 1/u$ and $f/f' = u\log u$: the answer is $4u\log u$, not $4\log u$. ADDED FOR CLARITY S006·13

Exam trap. In the second order use $g'(u)$, the derivative of g with respect to u, not $f'(u)$: for $g = \frac{1}{2}\tan u$, $g' = \frac{1}{2}\sec^2 u$. S006·17 · T006b 00:12:42

Worked examples

W4.1 · Three variables, sin u (slide 006·4) ESA ★★★

GIVEN $u = \sin^{-1}\dfrac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$; show $xu_x + yu_y + zu_z + 3\tan u = 0$.

STEP 1 $Z = \sin u = \dfrac{x(1+2y/x+3z/x)}{x^4\sqrt{1+(y/x)^8+(z/x)^8}} = x^{-3}g\left(\frac{y}{x}, \frac{z}{x}\right)$: degree -3.

STEP 2 Euler for Z: $xZ_x + yZ_y + zZ_z = -3Z = -3\sin u$.

STEP 3 $Z_x = \cos u\, u_x$ (and so on): $\cos u\,(xu_x + yu_y + zu_z) = -3\sin u$.

STEP 4 Divide by $\cos u$.

ANSWER $\boxed{xu_x + yu_y + zu_z = -3\tan u}$

CHECK sympy at two points: residual $< 10^{-14}$. Same as E4.1 with $f = \sin$, $n = -3$.

S006·4 · T006a 00:04:40 · T006a 00:08:22

W4.2 · Exponential: f = log (slides 006·12–13) ESA ★★★

GIVEN $u = e^{\frac{x^3y^3}{x^2+y^2}}$; find $xu_x + yu_y$.

STEP 1 $f(u) = \log u = \dfrac{x^3y^3}{x^2+y^2}$; $f(\lambda x, \lambda y) = \dfrac{\lambda^6 x^3y^3}{\lambda^2(x^2+y^2)} = \lambda^4 f$: $n = 4$.

STEP 2 $g = 4\dfrac{\log u}{1/u} = 4u\log u$.

STEP 3 Second order (added for clarity): $g' = 4(\log u + 1)$, so $g(g'-1) = 4u\log u\,(4\log u + 3)$.

ANSWER $\boxed{xu_x + yu_y = 4u\log u}$

CHECK sympy: both orders give residual 0.

S006·12 · S006·13 · T006b 00:07:59 · T006b 00:09:27

W4.3 · Logarithm: $f = e^u$ (slides 006·14–15) ESA ★★★

GIVEN $u = \log\dfrac{x^2+y^2}{x+y}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $f(u) = e^u = \dfrac{x^2+y^2}{x+y}$; $f(\lambda x, \lambda y) = \lambda f$: $n = 1$.

STEP 2 (i) $g = 1 \cdot \dfrac{e^u}{e^u} = 1$.

STEP 3 (ii) $g' = 0$: $g(g'-1) = 1\,(0-1)$.

ANSWER $\boxed{\text{(i) } 1,\ \text{(ii) } -1}$

CHECK sympy: (i) 1, (ii) -1 exactly.

S006·14 · S006·15 · T006b 00:09:42 · T006b 00:10:59

W4.4 · Inverse sine, degree ½ (slides 006·16–17) ESA ★★★

GIVEN $u = \sin^{-1}\dfrac{x+y}{\sqrt{x}+\sqrt{y}}$; find (i) $xu_x + yu_y$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy}$.

STEP 1 $f(u) = \sin u$; $f(\lambda x, \lambda y) = \dfrac{\lambda(x+y)}{\sqrt{\lambda}(\sqrt{x}+\sqrt{y})} = \lambda^{1/2}\sin u$: $n = \frac{1}{2}$.

STEP 2 (i) $g = \frac{1}{2}\dfrac{\sin u}{\cos u} = \frac{1}{2}\tan u$.

STEP 3 (ii) $g' = \frac{1}{2}\sec^2 u$: $g(g'-1) = \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right)$.

ANSWER $\boxed{\text{(i) } \frac{1}{2}\tan u,\ \text{(ii) } \frac{1}{2}\tan u\left(\frac{1}{2}\sec^2 u - 1\right) = \frac{1}{4}\tan u\,(\tan^2 u - 1)}$

CHECK sympy at two points: residuals $< 10^{-16}$. Equivalent form: $-\dfrac{\sin u\cos 2u}{4\cos^3 u}$.

S006·16 · S006·17 · T006b 00:11:22 · T006b 00:12:42

W4.5 · Inverse tangent (ESA) ESA ★★★

GIVEN $u = \tan^{-1}\dfrac{x^4+y^4}{xy}$. Prove (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(1 - 4\sin^2 u)$.

STEP 1 $f(u) = \tan u = \dfrac{x^4+y^4}{xy}$: degree $4 - 2 = 2$.

STEP 2 (i) $g = 2\dfrac{\tan u}{\sec^2 u} = 2\sin u\cos u = \sin 2u$.

STEP 3 (ii) $g' = 2\cos 2u$: $g(g'-1) = \sin 2u\,(2\cos 2u - 1) = \sin 2u\,(1 - 4\sin^2 u)$, using $\cos 2u = 1 - 2\sin^2 u$.

ANSWER $\boxed{\text{(i) } \sin 2u,\ \text{(ii) } \sin 2u\,(1 - 4\sin^2 u)}$

CHECK sympy: residuals 0 and $\approx 10^{-16}$.

ESA May-24 1a · ESA Jun-24 1b · CW U1 P20

Past-paper questions

ESA May 2024 (6 marks) UE22MA141A Q1.a · solved in W4.5

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 (6 marks) UE23MA141A Q1.b · solved in W4.5

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

SHORT-NOTE RECAP

If $f(u)$ has degree n: $g = nf/f'$, second order $g(g'-1)$. $e^{x^3y^3/(x^2+y^2)}$: $4u\log u$. $\log\frac{x^2+y^2}{x+y}$: 1, -1. $\sin^{-1}\frac{x+y}{\sqrt{x}+\sqrt{y}}$: $\frac{1}{2}\tan u$, $\frac{1}{2}\tan u(\frac{1}{2}\sec^2 u-1)$. $\sin^{-1}\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$: $-3\tan u$. ESA: $\tan^{-1}\frac{x^4+y^4}{xy}$ gives $\sin 2u$, $\sin 2u(1-4\sin^2 u)$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
λ	Scale factor	Replaces x, y by $\lambda x, \lambda y$ in the scaling test	E1.1
n	Degree	Of u (topic 3) or of $f(u)$ (topic 4)	E1.1–E4.2
$f(u)$	Homogeneous function of u	$\sin u$, $\tan u$, e^u, $\log u$	E4.1
$f(y/x)$ (slide 3)	Arbitrary function of the ratio	$Z = \sqrt{x}\, f(y/x)$; not the $f(u)$ of topic 4	W2.1
$g(u)$	Extension function	$nf(u)/f'(u)$; equals $xu_x + yu_y$	E4.1, E4.2
$g'(u)$	Derivative of g	With respect to u	E4.2
Z	Auxiliary function	$\cos u$ (slide 3) or $\sin u$ (slide 4, transcript)	W2.1, W4.1
z	Third variable	Lowercase, on slide 4 only	W4.1
$u_x,\ u_{xy}$	Partial derivatives	$\partial u/\partial x$, $\partial^2 u/\partial x \partial y$	E2.1, E3.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	E3.2

B Sources, coverage and gaps

Source	Path	Use
Deck 006 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.*	18 slides; content on 3–17
Transcript 006 part I (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.*	9.1 min: slides 3–4
Transcript 006 part II (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.*	12.9 min: slides 5–17
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problems 19–23 (with answers)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	6 in-scope Q1 parts
Videos 006 (2 .mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_*.mp4	Not opened: every slide is readable and every gap was settled by algebra

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler problems appear in six of them: Jul 2023 Q1.a (degree 5: $5u$, $20u$), May 2024 Q1.a and June 2024 Q1.b (the same $\tan^{-1}$ extension question), Jan 2024 set 3 Q1.c ($17\pi/2$), set 4 Q1.d ($9u$) and set 1 Q1.d (degree 0). Five are 6-mark questions. Testing homogeneity alone and verifying Euler's theorem by direct differentiation (topics 1–2) were not asked on their own; they appear only as the first step of a longer question.

Known gaps, conflicts and corrections

1. Slide 006·3 is titled "Verify Euler's theorem for $\cos^{-1}(\dots)$" but verifies it for $Z = \cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$, not for u. For u itself the result is $xu_x + yu_y = -\frac{1}{2}\cot u$ (added for clarity, checked with sympy).
2. Slide 006·3's middle factor is printed as $x^2\,\frac{1(1+y/x)}{1+\sqrt{y}/\sqrt{x}}$; the power is $x^{1/2}$, as the next step $\sqrt{x}\,f(y/x)$ shows.
3. Slide 006·4 prints the denominator as $\sqrt{x^8+y^8+z}$ in its solution line; it is $\sqrt{x^8+y^8+z^8}$, as in the question.
4. Slide 006·5's function is identically 1: $x^4+2x^2y^2+y^4=(x^2+y^2)^2$. The degree-0 answer is right, but the example is trivial; noted as a check.
5. Slide 006·7's function simplifies to $\frac{xy}{(x+y)^2}$ (added for clarity), which shows the degree 0 at a glance.
6. The lecture reads slide 006·8 as $e^x - e^{-y}$ (T006b 00:04:40); the slide has $e^x - e^y$. Either way the function is not homogeneous.

7. The lecture gives the degree on slide 006·16 as “plus 2 ... plus one by two” (T006b 00:12:00); the degree is $\frac{1}{2}$.
8. The lecture calls the second-order corollary $n(n-1)u$ an “extension of Euler's theorem” (T006b 00:05:55, 00:07:23). In this pack “corollary” means $n(n-1)u$ and “extension” means the $f(u)$ form $nf(u)/f'(u)$, as on slide 005·7.
9. The lecture says the answer to slide 006·9 (i) “becomes minus half ...” (T006b 00:06:59); the slide and the algebra give $-u = -\frac{x-y}{x^2+y^2}$ with no half.
10. Slides 006·12–13 ask only for the first-order sum; the second-order result $4u\log u\,(4\log u + 3)$ is added for clarity (sympy-checked).
11. Transcript part I ends with “Next we will solve some problems approximation” (T006a 00:08:53); that belongs to a later class and is not included.
12. The theory of class 005 is in its own pack; the statements used here are repeated without proofs.
13. Transcript recognition errors were corrected silently: “Sign in us” / “sine in was” → $\sin^{-1}$, “calc is” → cancelling, “tau” / “tanue” → $\tan u$, “second square” → $\sec^2$, “lambda power x power 3” → $\lambda^6 x^3 y^3$, “doux” → ∂x, “y'all” → $\sqrt{y}$.

Coverage

Source	Citations	Slides cited (slide × citations)
S006	68	3×4, 4×5, 5×7, 6×3, 7×2, 8×5, 9×1, 10×4, 11×5, 12×2, 13×8, 14×2, 15×8, 16×3, 17×9
CW	7	
ESA	11	
T006a	15	
T006b	33	

Slides 006·1, 006·2 and 006·18 are title, session and closing slides with no content. Every content slide (006·3–17) is worked in full.